THE OHIO STATE UNIVERSITY

May 2, 2025

RE: Letter of Support for an ASC Center of Excellence in Applied AI

Dear Joachim,

I enthusiastically support the creation of the BuckAI Observatory, an applied-AI Center of Excellence in the College of Arts and Sciences. The Observatory will lower technical barriers, catalyze interdisciplinary collaborations, and position OSU as a national leader in AI-driven Earth observation.

Research in my lab involves GNSS and other Earth observation data to model natural disasters as they are unfolding. We are currently exploring the use of AI in several ways, but the most relevant to BuckAI is the use of optical imagery and SAR observations with AI to determine the value of assets in tsunami inundation and flood zones for loss estimation. At present, this project has a small regional focus in the western United States, but our goal is to perform these risk and loss estimation calculations globally where validation datasets (such as building inventories) are not available. I expect that a seed investment by ASC in shared storage and computational resources, matched with initial technical support (a data engineer and student assistance with manually generating training data), as well as the other initiatives outlined in your white paper (such as championing interdisciplinary teams and annual workshops) will increase my success in pursuing extramural funding from NASA, FEMA, and various insurance/reinsurance agencies. Moreover, my collaborators have discussed potential commercialization of our loss estimation platform, and we would greatly benefit from the collective resources of BuckAI.

Beyond research, I endorse the proposed goal of better educating our students for a marketplace in which AI will play an increasingly critical role. I expect that students in my department will be interested in such classes (such as ES5757 “AI in Geosciences”) and plan to either share domain-specific case studies and/or contribute to new AI-related courses in my own unit.

Should ASC support the founding of the BuckAI Observatory, please reach out to me about how I can be involved and/or for any further information that would be helpful.

Sincerely,

Brendan Crowell
Assistant Professor
The Ohio State University
School of Earth Sciences